SOUTH DAKOTA STATE UNIVERSITY

AI-Powered Approach To Biomedical Dataset Discovery: Bridging Search Gaps In The NCBI Gene Expression Omnibus (GEO)

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What Is NCBI GEO

- National Center for Bioinformation Technology (NCBI)
- Gene Expression Omnibus (GEO)
  - Public Database
  - Functional genomics data
  - Over **1 million biological samples**
  - **~200,000 studies** spanning 2000–present
- Usage
  - Meta-analysis and validation
  - Biomarker discovery
  - Machine learning training data
  - Drug target identification

NCBI GEO Data Structures

- GSM (Biological Sample)
  - Atomic Unit
    - Single Biological Sample, One tissue biopsy, One cell line measurement etc
- **GSE (Series Records)**
  - Study level record – Group of samples from single experiments
  - Primary record for researcher
- GDS (Curated Datasets)
  - Analysis ready

MeSH (Medical Subject Headings)

- Curated biomedical vocabulary from National Library of Medicine(NLM)

  - 31000+ clinical terms
  - Updated once a year

- Powerful entry terms
  - Ex: Heart attack – Myocardial Infarction
  - Bridge between informal and formal scientific terminology in GEO meta data

How NCBI GEO – Search Works (Core Issue)

- Keyword search with Boolean operators
- Metadata field filtering (organism, platform, date)

Limitation	Example
Partial Synonym coverage	Heart attack vs myocardial infarction (terms in NCBI index).
No semantic understanding	“T-cell exhaustion” misses “dysfunctional CD8 cells”, “PD-1 high TILs”
No natural language queries	Researchers must write Boolean syntax, not ask questions
No biological knowledge	Genes, pathways, cell types treated as isolated text strings
No cross-dataset discovery	No “similar studies” or recommendation capability

Solution: AI Powered Discovery

- User query → (Ex: pancreatic cancer RNQ-Seq in humans)

Mesh Query Expansion →. (matched against MeSH 2026 vocab)

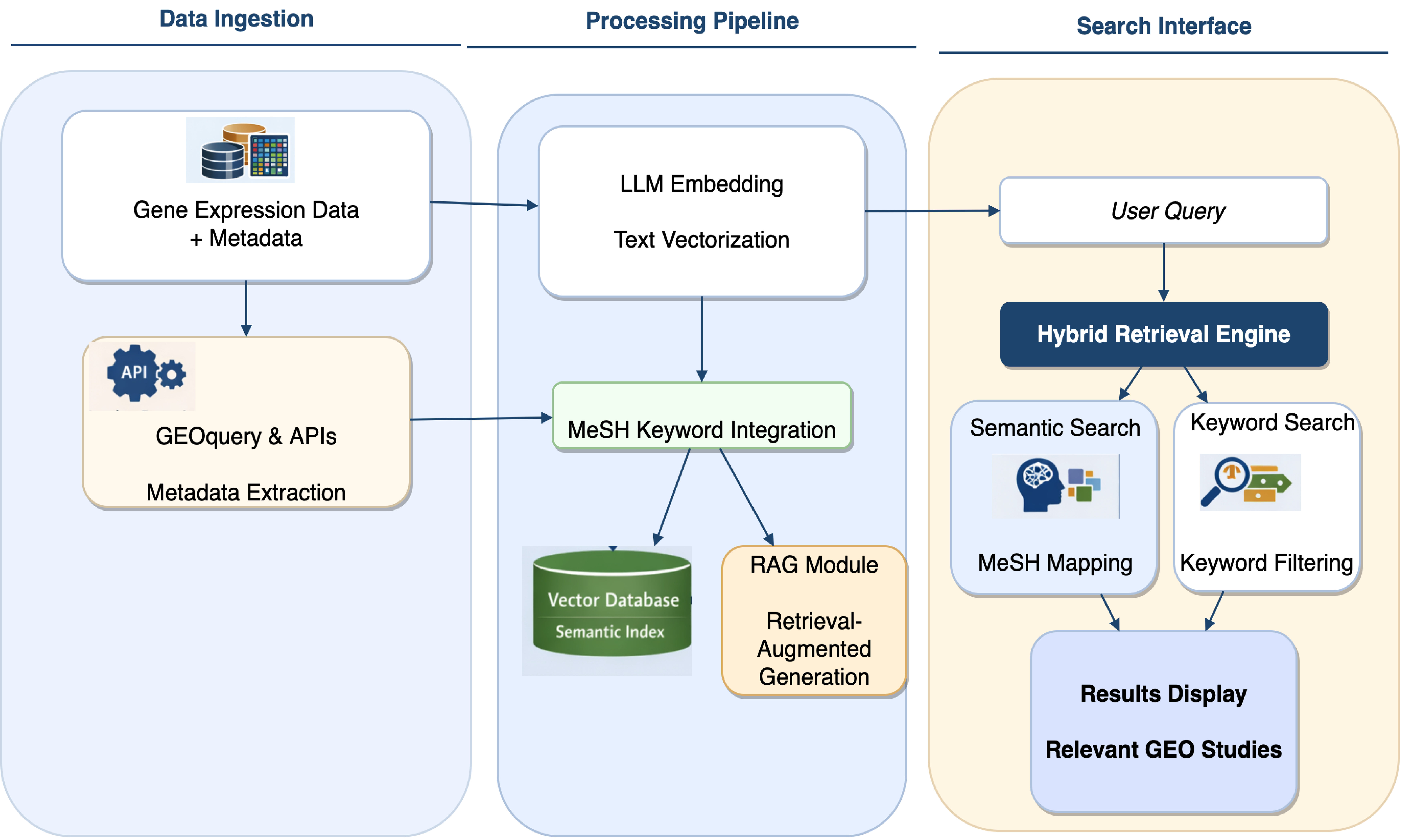
Dense Semantic Search (Vector db) → relevant medical concepts

Lexical Search (Exact Match for specific terms) → keyword search

Combine all 3 ->

Ranked and return the GEO Datasets (Ranked)

System Architecture



GEO Power Search – Features/Solutions

Challenge	GEOSearch Solution
Vocabulary mismatch	MeSH 2026 expansion (31,110 terms, 95% query coverage)
Missing semantic context	Dense vector search via Milvus
Poor metadata quality	Automatic MeSH tagging at ingest (100% coverage)
Keyword-only ranking	Reciprocal Rank Fusion (RRF) ⭐ multiple search methods fused into relevance
No natural language interface	RAG Q&A grounded in real GEO datasets ⭐
No downstream analysis path	iDEP integration for DE + pathway analysis

GEO Power Search – API Interface

GEO Smart Search - API 1.0.0 OAS 3.1

/openapi.json

AI-powered semantic search over NCBI GEO dataset metadata.

System

- GET `/health` Health
- GET `/stats` Stats

Search

- POST `/search` Search Post
- GET `/search` Search Get

Q&A

- POST `/ask` Ask

Datasets

- GET `/datasets/{accession}` Get Dataset

GEO Search - Semantic Search

- Convert text to Numbers , Words that mean the same end up close together in vector space

- Data
  - During ingest, GEO dataset (title, summary, design) converted to vector
  - During Search, Milvus finds the most similar by cosine similarity

- **Model used:** sentence-transformers/all-MiniLM-L6-v2
  - Trained on biomedical and general text
  - Runs locally, no API cost, ~1,000 texts/second on CPU

GEOSearch: Why RAG

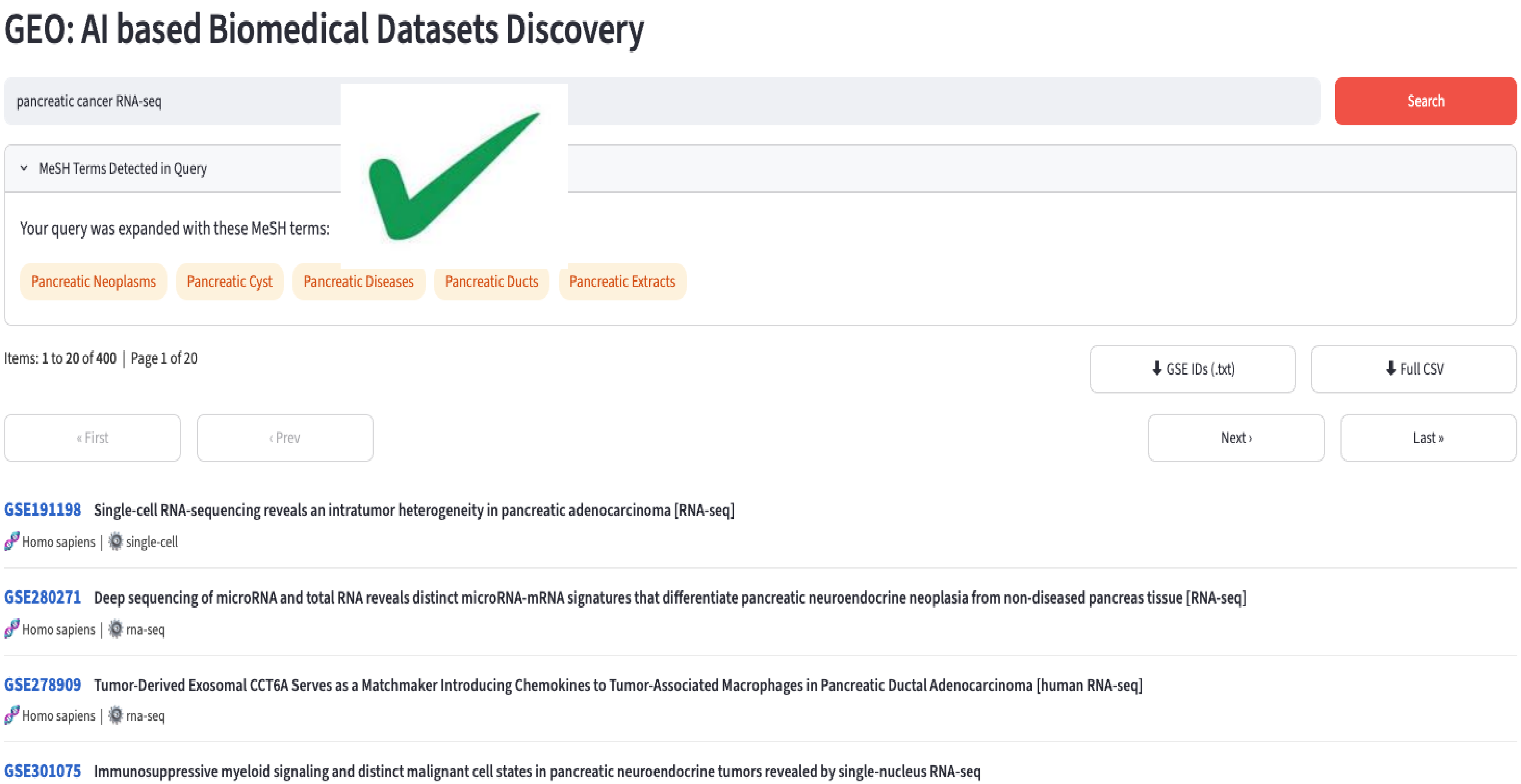
- **Without RAG (pure LLM):**
  - The model may hallucinate dataset names
  - Cannot know about recent GEO submissions
  - Answers are not traceable to real accessions

- **With GEOSearch RAG:**
  - Your question is used to retrieve the top-10 most relevant GEO datasets
  - Those datasets become the context given to the LLM
  - The LLM answers *only* using that context, citing real GSE accessions

GEO Search

Search Query: Pancreatic Cancer RNA-Seq

****** Mesh Terms detected/ download GSE id's for downstream analysis**



Search Query: heart attack (lay term)

****** Mesh Terms detected/ download GSE id's for downstream analysis**

GEO: AI based Biomedical Datasets Discovery

heart attack

Search

MeSH Terms Detected in Query

Your query was expanded with these MeSH terms:

Myocardial Infarction | Anterior Wall Myocardial Infarction | Inferior Wall Myocardial Infarction | ST Elevation Myocardial Infarction | Non-ST Elevated Myocardial Infarction

Items: 1 to 20 of 131 | Page 1 of 7

GSE IDs (.txt) | Full CSV

« First | ‹ Prev | Next › | Last »

GSE103182 Whole blood transcriptional profiling by RNA sequencing in patients with ST-elevation and non-ST elevation myocardial infarction

Homo sapiens | rna-seq

GSE254068 miR-619-5p and cardiogenic shock in patients with ST-Segment-Elevation Myocardial Infarction

Homo sapiens; synthetic construct | microarray

GSE213144 Different expression profiles and bioinformatics analysis of circular RNAs between ST-elevation myocardial infarction and chronic artery syndromes.

Homo sapiens

Search Query: T-cell exhaustion in tumor

****CD8 T-cell exhaustion, T-cell dysfunction. ** Semantic Search Win **

GEO: AI based Biomedical Datasets Discovery

T-cell exhaustion in tumor

Search

› MeSH Terms Detected in Query

Items: **1** to **20** of **82** | Page 1 of 5

⬇ GSE IDs (.txt)

⬇ Full CSV

GSE246651 Harnessing T-cel … :osis to isolate patient-derived tumor-specific TCR
Homo sapiens

GSE141787 CD8+ T-cell exhaustion induced by leukemic cells drives progression in Chronic Lymphocytic Leukemia
Homo sapiens

GSE227316 PGE2 effect on unstimulated and exhausted human CD8+ T-cells
Homo sapiens

GSE166213 Tumor methionine metabolites reprograms chromatin accessibilities of CD8+ T cells linking to T cell dysfunction.
Homo sapiens | atac-seq

GSE145809 Single Cell Transcriptomic Characterization of CAR T-Cell Products Reveals Subpopulations, Stimulation and Exhaustion Signatures
Homo sapiens | single-cell